

Seokjun Ryu

✉ seory0@outlook.kr • 🌐 seory0.github.io • 🐙 SEORY0 • in seory0

Research Interests

Offensive security researcher and CTF player focused on **AI Security**, **Windows Kernel Exploitation**, and **Reverse Engineering**, with vulnerability discovery across **AI/ML frameworks and AI agent platforms** (vLLM, OpenClaw, Flowise) and **LLM-based multi-agent systems** for automated vulnerability research.

Education

Soongsil University

B.S. in Electronic Engineering (IT Convergence)

Seoul, South Korea

Mar. 2022 – Present

KITRI WhiteHat School

3rd Cohort, Vulnerability Analysis Track (Completed)

Mar. 2025 – Sep. 2025

○ Project: Windows kernel driver 1-day analysis and exploitation (CVE-2024-35250)

POC Security

Hacking Camp 29th (Completed)

Aug. 2024

Work Experience

AI Safety Center, Soongsil University

Undergraduate Research Intern, AI for Security

Seoul, South Korea

Jan. 2026 – Present

Publications

Domestic Conference

Analysis of Vulnerability Attack Pathways in AI/ML Frameworks Based on Public CVEs and a Lightweight LLM Prompting-Based Bug Hunting Methodology

May 2026

Korea Institute of Information Security & Cryptology Summer Conference 2026 (Best Paper Award)

Seokjun Ryu, Sohee Park, Domin Ok, and Daeseon Choi

Talks

AI for Security for AI: Hacking AI Agents with AI Agents

Aug. 2026

Hacking Camp 33rd, POC Security

Seokjun Ryu and Taeyoung Jeong

Vulnerability Disclosures

CVE-2026-62203 OpenClaw

Disclosed via GHSA

CVE-2026-35626 OpenClaw

Disclosed via GHSA

CVE-2026-34755 vLLM

Disclosed via Huntr

GHSA-c2p8-**-****** OpenClaw

Disclosed via GHSA, CVE pending

GHSA-rwm5-**-****** OpenClaw

Disclosed via GHSA, CVE pending

GHSA-rj8f-**-****** OpenClaw

Disclosed via GHSA, CVE pending

GHSA-7cp7-**-****** OpenClaw

Disclosed via GHSA, CVE pending

GHSA-5968-**-****** Flowise

Disclosed via GHSA, CVE pending

Projects

- Development of Multi-Agent Vulnerability Research SystemMay 2026 – Aug. 2026
- Achieved **91.0% pass@1** on CyberGym Level 1 (1,372/1,507 tasks) by building **mneme**, a multi-agent system with structured causal memory and execution-feedback refinement, matching Microsoft MDASH and beating the 66.6% same-model (Claude Opus 4.6) baseline by 24.4 points. [Report] [CyberGym] [Press]
 - Directed the Mythos+ Task Force as **TF Lead**, owning operations, agent architecture design, and the evaluation framework.

- Development of Robust AI and Distributed Attack Detection Technologies for Edge AI SecurityMar. 2026 – Present
- Built the **frontend and backend** of an IITP-funded Edge AI Vulnerability Analysis System prototype, a unified dashboard spanning adversarial-example, adversarial-patch, and voice attacks across image-classification, object-detection, and speech-recognition models.
 - Automated model/attack selection, simulated attack execution, on-demand attack-data generation, and result reporting.

- Automated Agent-Targeted Bug Hunting with AI Agents (Agent-Zero)Mar. 2026 – Aug. 2026
- Discovered **38 valid vulnerabilities** across OpenClaw, n8n, and qbee-agent by building **CrayFisher**, an agentic zero-day bug-hunting agent based on an OpenClaw 1-day vulnerability taxonomy targeting AI-agent codebases.
 - Led the project as **Team Lead** (ASC hacking club).

- Windows Kernel Privilege-Escalation 1-Day Analysis and Exploitation (CVE-2024-35250)Mar. 2025 – Sep. 2025
- Developed a working local privilege-escalation exploit for **CVE-2024-35250**, gaining a SYSTEM shell by building a kernel read/write primitive via PreviousMode overwrite under KASLR/SMEP/kCFG and overwriting the process token.
 - Identified the root cause with IDA and WinDbg: a kernel streaming driver re-issues an IOCTL with the new IRP's RequestorMode hardcoded to KernelMode, invoking a user-supplied function pointer without validation (trust-boundary violation).
 - Led **root-cause analysis and technical documentation** at WhiteHat School (KITRI). *Stack: C++, Assembly.*

Honors and Awards

- Awards
- Best Paper Award (National Security Research Institute Director's Award)May 2026
- Korea Institute of Information Security & Cryptology Summer Conference 2026

- CTF Competitions
- Teams: ASC (ssu-asc.com) • Fermion (project-plasma.org) • Physical Lab (phylab.space)
- Cyber Conflict Exercise 2026 Team 클로드짱의 성능은 망가져있다Finalist (6th)
- Hacksium Busan 2026 Team 나는야임영웅Finalist
- Hacktheon International University Student Cyber Security Competition 2026 Team 나는야임영웅Finalist
- HyperSonic CTF Team ASC2nd
- BITSCTF 2026 Team Fermion4th
- HSpace Club League 25 Team ASC5th

Skills

- Programming: C, C++, Python, Assembly
- Security: Windows Kernel Exploitation, Reverse Engineering, Vulnerability Research, AI Agent Security